MG Road Core - MG-ACC-38 Arista Access Node QoS Congestion Event Root Cause Isolation and Resolution SOP

Node: MG-ACC-38 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_6002

Description

The MG Road Core - MG-ACC-38 Arista Access Node QoS Congestion Event is a critical failure domain that affects the entire access network, impacting the overall Quality of Service (QoS) and causing microbursts, interface tail drops, and QoS queue starvation. This failure domain occurs on the Arista Access Node due to the complex interplay of QoS policies, traffic shaping, and policing mechanisms implemented within the node. The root cause of this failure domain lies in the misconfiguration of QoS policies, which can lead to inadequate resource allocation and inefficient traffic management. Furthermore, the Arista Access Node's hardware capabilities and software limitations can exacerbate the issue by inducing packet loss, latency, and jitter. To isolate and resolve this failure domain, it is crucial to employ a structured approach that combines thorough diagnostics with precise resolution steps, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Verify that the MG Road Core site is accessible and safe for remote hands to perform the necessary checks and repairs.

- Confirm that the redundant link status is operational and ready to take over in case of a primary link failure.

- Ensure that all necessary permissions and access credentials are obtained before initiating the troubleshooting process.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Inadequate resource allocation	Medium
Hardware limitations	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy all, Expected Output: QoS policy configuration and statistics, Description: This command displays the QoS policy configuration and statistics, allowing the identification of misconfigured policies and potential resource allocation issues.
```

```
Command: show interface statistics, Expected Output: Interface statistics, including packet loss and latency, Description: This command provides interface statistics, including packet loss and latency, which can help isolate the root cause of the congestion event.

Command: show traffic policer all, Expected Output: Traffic policer configuration and statistics, Description: This command displays traffic policer configuration and statistics, enabling the identification of potential traffic management issues.
```

Resolution Steps

Step 1: Step: Reconfigure QoS policies, Command: config qos policy (set to desired value), Description: Reconfigure QoS policies to ensure adequate resource allocation and efficient traffic management.

Step 2: Step: Adjust resource allocation, Command: config resource allocation (set to desired value), Description: Adjust resource allocation to ensure sufficient resources are available for optimal network performance.

Step 3: Step: Upgrade hardware, Command: upgrade (set to desired value), Description: Upgrade hardware to eliminate hardware limitations and ensure optimal network performance.

Post-Check Verification

```
Command: show qos policy all, Expected Output: QoS policy configuration and statistics, Description: Verify that QoS policies are correctly configured and statistics are within expected ranges.

Command: show interface statistics, Expected Output: Interface statistics, including packet loss and latency, Description: Verify that interface statistics, including packet loss and latency, are within expected ranges.

Command: show traffic policer all, Expected Output: Traffic policer configuration and statistics, Description: Verify that traffic policer configuration and statistics are within expected ranges.
```

Roll-back Procedures

- Step: Restore original QoS policies, Command: config qos policy (restore original value), Description: Restore original QoS policies to ensure minimal downtime and optimal network performance.

- Step: Revert resource allocation, Command: config resource allocation (revert original value), Description: Revert resource allocation to ensure minimal downtime and optimal network performance.

- Step: Downgrade hardware, Command: downgrade (revert original value), Description: Downgrade hardware to eliminate hardware limitations and ensure minimal downtime and optimal network performance.